through the opening, you'd see the inside of the power plant, the location of the second mouth, as in Figure 15.2b. Another misleading feature of Figure 15.2a is that the wormhole doesn't appear to be a shortcut. We can fix this by modifying the illustration as in Figure 15.3. As you can see, the usual route from the power plant to the Kwik-E-Mart is indeed longer than the wormhole's new spatial passage. The contortions in Figure 15.3 reflect the difficulties in drawing general relativistic geometry on a page, but the figure does give an intuitive sense of the new connection a wormhole would provide.

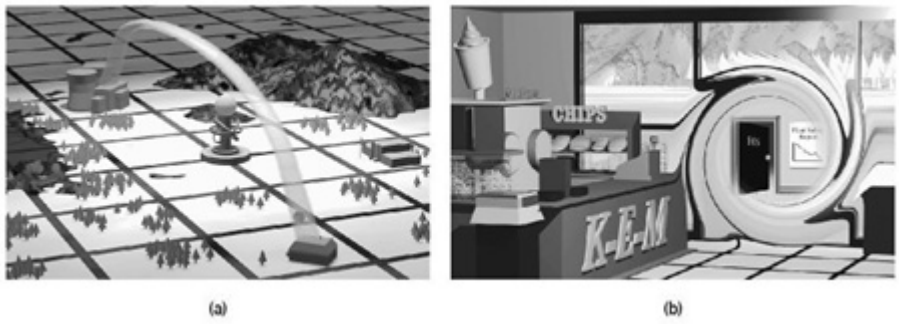


Figure 15.2 (a) A wormhole extending from the Kwik-E-Mart to the nuclear power plant. (b) The view through the wormhole, looking from the mouth at the Kwik-E-Mart and into the mouth in the power plant.

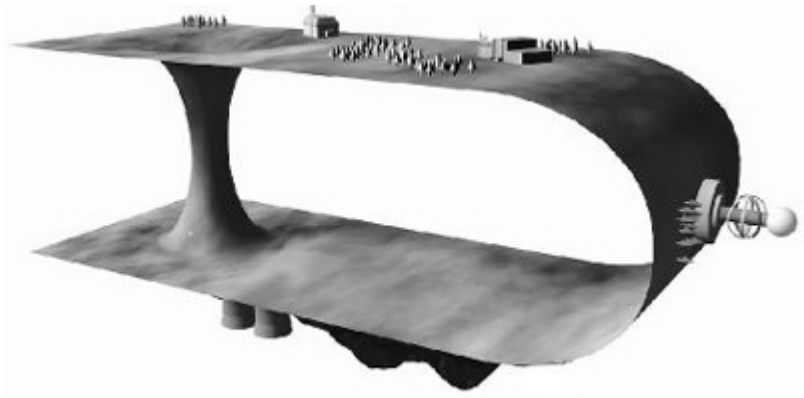


Figure 15.3 Geometry which more clearly shows that the wormhole is a shortcut. (Wormhole mouths are really inside Kwik-E-Mart and the nuclear power plant, although that is difficult to show in this representation.)

No one knows whether wormholes exist, but many decades ago physicists established that they are allowed by the mathematics of general relativity and so are fair game for theoretical study. In the 1950s,